A sharper certified contraction bound for 3-SAT

Rectangular multiplication, coupled support certificates,
and the remaining logarithmic barrier

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26 October 2026 (Pacific time)

Abstract

We review the 3-SAT research at a fixed ProveIt revision, including all five new incoming archives absent from its 29-packet index. The strongest comparable generic cached contraction coefficient is $29.81160894538347\ldots$, multiplying $s/\sqrt{\lceil \log_2 s \rceil}$ in a base-two exponent. A weighted triple certificate lowers that coefficient to $29.77681894508374\ldots$ with the same matrix backend and the same recursion schedule. Incorporating newer, explicitly identified characteristic-zero matrix-multiplication results and a newly certified schedule gives coefficients $28.43373937834166\ldots$ for time and $28.30578003029975\ldots$ for space. Every recursive table entry is charged. We prove a finite-base arithmetic transfer that permits the imported matrix algorithms over the two-adic coefficient rings without a map from those rings to odd prime fields. We also prove a restricted rectangular-grid minimax obstruction for the Anthropic crossing recursion and supply exact finite operator-rank probes. These results improve explicit upper-bound coefficients. They do not prove a smaller general 3-SAT exponent order than $n \log\log n/\sqrt{\log n}$, or an asymptotic improvement over the unspecified constants of the independent Anthropic algorithm. No end-to-end SAT implementation or new Lean formalization is claimed.

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1 Outcome and exact scope

The objective was an improvement for *3-SAT*, with a smaller logarithmic order preferred to another constant improvement. The strongest result established in this investigation is a strict improvement of the explicit generic contraction certificate. The stronger objective remains unresolved.

Write

$$\lambda = \lceil \log_2 s \rceil, \qquad \kappa = \frac{\pi\sqrt{2}}{\ln 2}.$$

Here s is a normalized Boolean wire budget, *not* the number of SAT variables. The common interface and all its entry costs are stated in Section 3.

Theorem 1.1 (Certified improvement)**.** *For the generic input interface of Section 3, there is a deterministic exact contractor with*

$$T(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)} 2^{(C_T+o(1))s/\sqrt{\lambda}}, \qquad S(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)} 2^{(C_S+o(1))s/\sqrt{\lambda}},$$

where $C_T = 28.43373937834166\ldots < 28.434$ *and* $C_S = 28.30578003029975\ldots < 28.306$ *are defined exactly in Section 6. Thus* 28.434 *and* 28.306 *are safe eventual coefficients for time and space, respectively. The theorem imports the explicitly listed Cauchy deformation, mode, Hermite-recovery, packing and entry-representation results, and the characteristic-zero rectangular matrix-multiplication theorem of Uznański [18]. The new finite certificates and complete schedule are proved and checked here. The displayed space bound is subexponential.*

Independently of the newer matrix theorem, the weighted triple certificate with the preceding packet's 19/8 *and* 3/10 *backend and its unchanged schedule gives time and space coefficient* $29.77681894508374\ldots$.

The comparison in this theorem is between *proved upper-bound functions under the same normalized interface*. It is not a measurement of SAT running times. In particular, replacing C_{old} by $C_{\text{new}} < C_{\text{old}}$ changes the certified leading envelope by

$$2^{-(C_{\text{old}}-C_{\text{new}}+o(1))s/\sqrt{\lambda}}.$$

This is a strict asymptotic improvement in that envelope. It does not establish a ratio between the actual running times of two implementations.

1.1 What transfers to 3-SAT

The source's direct clause/equality encoding at bounded clause density has $s = O(n)$, while its general reduction has $s = O(n \log\log n)$. Substitution therefore retains

$$\operatorname{poly}(n+m)\, 2^{O(n/\sqrt{\log n})} \quad (m \le Dn,\ D \text{ fixed}), \qquad \operatorname{poly}(n+m)\, 2^{O(n\log\log n/\sqrt{\log n})} \quad \text{in general.} \tag{1.1}$$

The inherited identities count satisfying assignments, so testing whether the resulting count is positive decides 3-SAT. The mathematical contractor is more general because that is the interface used by the 3-SAT reduction; no other SAT width or unrelated counting problem is pursued here.

The reviewed papers do not give a common explicit conversion from every generic coefficient per normalized wire to a coefficient per variable for the dense reduction. Also, the Anthropic algorithm has a different architecture and unspecified constants in its O notation. Hence a claim that $28.43373937834166\ldots$ is a universal best constant for 3-SAT, or that this algorithm asymptotically beats the Anthropic algorithm on all inputs, would be unsupported.

1.2 What is new, inherited, and still missing

The new work consists of the complete weighted-triple certificate, its strict improvement with the old backend, the finite and exact use of the updated backend, the new paid schedule, the rectangular-grid minimax calculation, and executable rank probes of the actual crossing operators. The incoming pair-certificate report explicitly suggested higher-order support certificates as future work; our triple result completes one concrete instance of that suggestion. We make no priority claim for quadratic identities, averaging, matrix interpolation, field descent, or the general idea of a certificate hierarchy.

The logarithmic-order improvement would require a new structural theorem. Section 7 records what the examined approaches can prove, why several plausible substitutions fail, and a precise crossing-factor target that would suffice. Failed approaches and finite experiments are kept separate from the contractor theorem.

2 Frozen source review and the actual comparison frontier

2.1 Revision and review depth

The ProveIt review is frozen at

`8e76d515063c4b9fbd8743a846b457e3a01bc039.`

The relevant recent commits are summarized below. Times in the source metadata are UTC; these changes occurred during the evening of 26 October in Pacific time. Source hashes and exact paths are recorded in the archive.

Commit prefix	UTC time, 27 Oct.	Relevant change
`6d3adc4506`	02:03	Eight round-2 packets placed.
`450252e9d8`	02:36	Five round-3 packets placed; omitted scripts restored.
`a7ae5ab55f`	03:15	Nine round-4 packets placed.
`7de9d09fec`	03:20	Public Anthropic Lean development linked.
`26bd83d4b3`	03:51	Five new affine/pairwise ZIP archives arrived.

The project index contains 29 refinement packets: seven in round 1, eight in round 2, five in round 3, and nine in round 4. The five newest incoming archives raise the reviewed inventory to 34. We screened all indexed claims and read the full mathematical sections that determine the current frontier, inherited interfaces, and proposed order improvements. This is not an assertion of an independent end-to-end proof audit of every earlier packet. The inventory states which source claims are being summarized; the main theorem has a narrower explicit dependency list.

During completion, commit `324f3d9b009a` placed the five incoming packets into the project. All 213 placed files matched the reviewed ZIP members by Git blob identity, including all 160 LaTeX, Python and JSON files; the omitted files were documented ancillary or build files. The closing observed main revision was `0cf10036e6ad`. This placement changes the index to 34 packets without changing the reviewed mathematics. The full identity check accompanies the archive.

2.2 The five archives missing from the index

All entries below concern the same generic normalized input model and multiply $s/\sqrt{\lambda}$ in a base-two exponent. They are source claims whose finite frontier certificates were inspected; the two leading pairwise checkers were also replayed.

Incoming archive stem	Time coefficient	Space coefficient
`sat_pairwise_certificates`	29.811609	29.811609
`sat_affine_contraction`	30.257036	25.989068
`sat_affine_geometry`	30.269873	29.487537
`sat_affine_cauchy`	30.413349	29.466030
`sat_affine_plane`	30.691338	29.466030

The first two rows form the relevant time–space tradeoff among these five. A smaller time coefficient does not erase the second row's smaller memory coefficient. The best generic polynomial-space coefficient remains 46.184845586731... from `sat_adaptive_bounds`. The newest pairwise packet separately gives 16.812233801883... in polynomial space at root degree at most $\lceil s^{1/12} \rceil$. That lower-degree result has stronger assumptions and is not a competitor in the generic degree-$s^{9/10}$ table.

The older indexed generic cached record was 30.891018920366.... Comparing a new result only with that value would miss the incoming 29.8116... frontier. None of the 34 packets establishes a smaller general order than (1.1). The conditional improvements require an additional algorithm or representation theorem that is not provided by the corresponding packet.

2.3 The two primary algorithms and formal verification

The vendored OpenAI manuscript at upstream revision `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a` proves the bounds in (1.1), with polynomial space, in its quantitative appendix [1]. Its published Lean scope page reports the weaker general exponent $n(\log\log n)^2/\sqrt{\log n}$ for that formal development [2]. The paper's sharper quantitative appendix and its formalized theorem should not be conflated.

The Anthropic source is inspected at the user's exact commit

`1a6eee293b9e00cb432eef6018a21f523623a9af`.

Its `MainResults.lean` states a finite-alphabet deterministic Mathlib Turing-machine counting theorem, and `Solution.lean` supplies proofs without extra hypotheses. Specializing to 3-SAT gives

$$2^{O\left(\min\left\{\frac{m}{\sqrt{\log m}}, \frac{n\log\log n}{\sqrt{\log n}}\right\}\right)}. \tag{2.1}$$

The stated counting form assumes normalized clauses and occurring named variables; small cases and unused variables are handled by the usual front end. Raw input normalization adds a polynomial factor in input length. The separate ETH wrapper is also proved [15].

We retrieved the exact-commit upstream verification log. The `verify (eth)` job reports successful Lean-kernel, nanoda, and Comparator acceptance, with the deliberately unproved challenge statements kept separate from `Solution.lean` [16]. This is observed upstream verification, not a local rebuild performed here. The older round-2 statement that the public development was unavailable is therefore obsolete. The Anthropic development does not formalize the ProveIt refinement packets or the new results in this report, and its displayed time theorem does not supply the OpenAI paper's polynomial-space claim.

2.4 A stronger available matrix backend

The newest pairwise packet still uses $\omega < 19/8$ and dual exponent $\alpha > 3/10$. Current primary sources contain stronger results. The OpenAI scope page records $\omega(\mathbb{C}) \le 9/4$ and $\alpha > 0.465$ [17].

Uznański's version-2 theorem gives $\omega(1, 1/2, 1) = 2$ and $\omega \le 9/4$ over characteristic zero, with positive exponent slack understood [18]. We use these two fixed shapes. This avoids assuming uniform constants for a continuously varying family of rectangular shapes. The transfer to the exact modular calculation is proved in Section 4.

3 The exact contraction interface

3.1 Inputs and coefficient rings

The input is a bipartite Boolean tensor network with at most s wires, polynomially many vertices, maximum degree at most $\left\lceil s^{9/10} \right\rceil$, and graph/tensor description length $\mathcal{L}$. Its value is

$$Z = \sum_{x \in \{0,1\}^E} \prod_v T_v(x|_{\delta(v)}).$$

Initial entries lie in $(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell)$ with $1 \le P_0, \ell \le s$. A uniform entry routine costs $(s + \mathcal{L})^{O(1)} 2^{O(s^{9/10})}$ time and polynomial space; the constants in the input promise are fixed.

The original graph and tensor descriptions are stored once. Generated tensors retain immutable references to those descriptions and to earlier generated descriptors, rather than copying or recursively expanding them. A reference can be represented by a descriptor index and input offsets. Random access is convenient but is not an extra input-model assumption: on a sequential-input machine, resolving an offset by scanning the original description costs $O(\mathcal{L})$ per access. That cost, and the fixed polynomial factor in the promised original entry routine, are charged to each access event. They do not multiply the number of recursive calls.

All allocation counts, mode-index loops, generated ring arithmetic, and descriptor-construction loops are controlled by s and the fixed menus; their iteration counts do not depend on $\mathcal{L}$. Source offsets use $O(\log(s + \mathcal{L}))$ bits; generated indices and memory addresses have polynomially many bits in s in addition to this offset cost. Consequently one fixed polynomial $P(s + \mathcal{L})$ bounds the description-access and address-processing cost per event throughout the recursion. We count work after factoring out this one polynomial: an original entry event still has weight $2^{O(s^{9/10})}$, and all subsequent generated-entry work is charged explicitly. If $\widehat{T}_d$ denotes this normalized work, then actual time is at most $P(s + \mathcal{L})\widehat{T}_d$. The polynomial is paid once in this comparison, not once per recursion depth. This is an implementation of the inherited immutable-entry interface, with no stronger bound on the input description length.

The inherited algorithm uses

$$R_p = (\mathbb{Z}/2^p\mathbb{Z})[U, V]/(f(U), V^\ell),$$

where f is monic of degree $O(\log s)$ with irreducible reduction modulo two. Rank over $\mathbb{Z}/2^p\mathbb{Z}$, precision, and the coordinate multiplication table have polynomial descriptions. Each tensor is tagged with its *creation precision*. A later entry evaluation first recovers its value at that precision and uses the same canonical coordinate representatives. In particular, no proof here treats canonical lifting as a ring homomorphism.

We import the Cauchy deformation and exact Hermite recovery from the quantitative appendix, the sharper Hermite precision from the precision refinements, and the asymmetric packing and joint Cauchy modes from the subsequent packets [1, 5, 6, 7, 8, 10]. The new algorithm changes the contraction of the resulting finite mode identity. It does not change its validity or its ring semantics.

3.2 Modes, entropy, and recursive tables

At an active call let M be the number of wires, r a fixed group alphabet, a the packet exponent, u the retained taper range, and $i < u$ a node valuation. Set $D = 2^{a-i}$. The compatible inherited bounds are

$$p_t \le (6 + o(1))MD/u, \qquad M_{\text{child}} \le (2/r + o(1))M. \tag{3.1}$$

On each directed group pair, there are at most $(2/r^2 + o(1))M/D$ mode families of each index j, with weights $a + 1 - i + 4j - 2\,\mathrm{popcount}(j)$.

Enumerate exact pair-weight allocations $(b_{gh})_{g\neq h}$ whose sum is less than p_t. For a fixed allocation let $\mathcal{I}_{gh}$ be the finite list of mode tuples of that exact weight, and skip allocations with an empty list. Define

$$h_{gh} = \log_2 |\mathcal{I}_{gh}|, \qquad \mathcal{K}(M,u) = \kappa M/\sqrt{u}.$$

The contribution of the allocation is exactly

$$\sum_{\xi_{gh}\in\mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e\ni g}). \tag{3.2}$$

Each entry of Z_g is a genuine smaller recursive contraction. It is not a free oracle or an already available table.

For every fixed nonnegative directed weight vector z, the joint-mode entropy estimate is

$$\sum_{g\neq h} z_{gh}h_{gh} \le \left(\frac{\|z\|_2}{r} + o(1)\right)\mathcal{K}(M,u). \tag{3.3}$$

For orientation, its leading constant follows from the logarithm of the mode generating product,

$$\sum_{j\ge 0} -\ln(1 - e^{-\theta(1+4j-2\,\mathrm{popcount}(j))}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)).$$

The common budget bounds the weighted natural logarithm by

$$\theta p_t + \frac{\pi^2}{24\theta}\frac{(2+o(1))M}{r^2 D}\,\|z\|_2^2 + o(M/\sqrt{u}).$$

Optimizing θ and using (3.1) gives (3.3). The inherited additive precision buffer makes the error uniform over the finite support family used below.

For unordered pairs write $g_{\{a,b\}} = h_{ab} + h_{ba}$. Then

$$v \cdot g \le \left(\frac{\sqrt{2}\,\|v\|_2}{r} + o(1)\right)\mathcal{K}(M,u). \tag{3.4}$$

The factor $\sqrt{2}$ is required because each unordered coefficient appears on both directed pairs. The corresponding uniform norm bound is $\|g\|_2 \le (\sqrt{2}/r + o(1))\mathcal{K}$, obtained by a finite-net argument in this fixed dimension.

The number of exact allocations is polynomial because r is fixed and all budgets are polynomial in s. A standard unbounded-knapsack recurrence for the mode weights computes each exact list cardinality. The same table supports rank/unrank and polynomial-delay enumeration. Thus the selector later used in the proof needs no counting oracle.

3.3 Packing and the recurrence that must be paid

With $L = 2^a$, $Q = r^a$, and side-degree bounds R, C, sufficient packing conditions are

$$QR/M = o(1), \qquad Q^2C(\log s)^2/M = o(1), \qquad LQ^2/M = o(1). \tag{3.5}$$

At the root both degrees are at most $s^{9/10+o(1)}$. At later calls the retained side stays below that bound and the fresh packet side has degree at most $2^{a_{d-1}+1}$.

A group has $2(r-1)$ incident directed pair indices. Forming its full table therefore requires at most $2^{(\alpha_r+o(1))\mathcal{K}}$ recursive entry evaluations, where

$$\alpha_r = \frac{\sqrt{2(r-1)}}{r}. \tag{3.6}$$

If a complete local contraction plan has coefficient β_r, the charged recurrence is

$$\widehat{T}_d \le \mathrm{poly}(s)\left(2^{(\alpha_{r_d}+o(1))\mathcal{K}_d}\widehat{T}_{d+1} + 2^{(\beta_{r_d}+o(1))\mathcal{K}_d}\right). \tag{3.7}$$

The first term includes all singleton tables and their recursively computed entries. The second term includes all subsequent merges and the last triangle. Improving the second term alone need not improve the maximum after unrolling. This distinction explains why a new schedule is necessary when the matrix backend improves.

4 A uniform exact rectangular backend

4.1 Two fixed shapes suffice

Fix $\varepsilon = 10^{-5}$. The imported characteristic-zero results give, with any fixed positive slack, square exponent $9/4$ and exponent two for the shape $N \times N^{1/2}$ by $N^{1/2} \times N$ [18]. We do not assume that an exponent infimum is attained by one finite algorithm with zero slack.

Lemma 4.1 (Uniform endpoint interpolation)**.** *For sorted nonnegative logarithmic dimensions $x \ge y \ge z$, the product of matrices with those three index dimensions can be computed, up to polynomial bit overhead and fixed rounding factors, with exponent*

$$F(x,y,z) = \max\left\{(1+\varepsilon/2)(x+y), \quad x + (3/4+\varepsilon)y + \tfrac{1}{2}z\right\}. \tag{4.1}$$

A fixed finite library of algorithms suffices for all shapes occurring in the contractor. Working storage is a polynomial factor times the largest input/output array, whose logarithmic size is $x+y$.

Proof. Choose finite bilinear endpoint algorithms with fixed positive slacks at most ε in their exponents. First block the largest dimension, paying $x-y$. If $z \ge y/2$, put

$$t_0 = 2(y-z), \qquad t_1 = 2z - y.$$

Both are nonnegative, $t_0 + t_1 = y$, and $t_0/2 + t_1 = z$. Tensoring thin and square algorithms gives exponent at most

$$x - y + (2+\varepsilon)t_0 + (9/4+\varepsilon)t_1 = x + (3/4+\varepsilon)y + z/2.$$

If $z < y/2$, pad the inner dimension to $y/2$. The exponent is at most $x + y + \varepsilon y \le (1+\varepsilon/2)(x+y)$. This proves the cost bound.

Only two endpoint shapes are used. Choose constant base dimensions compatible with the rational thin aspect ratio, and round the number of tensor iterations upward. Each rounding inflates all dimensions by fixed factors. If necessary allocate a smaller fixed slack to the endpoint choices to absorb these fixed overheads. No algorithm chosen from a continuum of input-dependent shapes is required.

Execute the fixed bilinear recursions depth first, processing one child product at a time and accumulating into output arrays. At every recursive level, the required linear forms and output arrays are a fixed number of arrays at that level's dimensions. Their storage sums geometrically along an active path. Blocking is also sequential. Thus storage is a constant or logarithmic factor times the largest input/output array. Permuting the three tensor legs justifies sorting the dimensions. □

For comparison, using only the formal-scope bounds $\omega(\mathbb{C}) \le 9/4$ and $\alpha > 0.465$ permits the conservative thin ratio $23/50$ and gives

$$\max\{(1+\varepsilon/2)(x+y),\quad x+(85/108+\varepsilon)y+(25/54)z\}.$$

The numerical headline uses the stronger half-width endpoint explicitly cited above; the weaker expression is recorded to make this dependency visible.

4.2 From complex identities to exact modular computation

The arithmetic model in the source matrix theorems is not itself a bit-complexity theorem for R_p. The following transfer closes that gap.

Lemma 4.2 (Fixed algebraic coefficients)**.** *Fix a finite matrix-multiplication bilinear algorithm that exists over $\mathbb{C}$. An algorithm with the same permitted bilinear support and operation pattern, and no larger rank, exists with coefficients in a fixed number field. Outside a fixed finite set of primes, its identities hold in a fixed-rank finite algebra over the prime field. These constants and this exceptional set are independent of the input matrix dimensions obtained by recursively composing the base algorithm.*

Proof. Correctness of the fixed bilinear formula is a finite system of polynomial equations with rational coefficients in the algorithm's constants: equate the coefficient of every input monomial to the required multiplication coefficient. Existence of a complex solution means the ideal is proper. By the weak Nullstellensatz over $\overline{\mathbb{Q}}$, it has an algebraic-number solution. The finitely many coordinates lie in one finite number field K. Fix an integral primitive element θ, a monic integer minimal polynomial g, and an integer $D > 0$ such that all formula constants lie in $\mathbb{Z}[\theta, 1/D]$.

For a prime $q \nmid D$, reduce the coefficients in

$$A_q = \mathbb{F}_q[T]/(g(T)).$$

The bilinear identity holds there because it is a polynomial identity. The polynomial g need not remain irreducible or separable modulo q: the computation is division free after the fixed scalar denominators have been inverted. The scalar inclusion $\mathbb{F}_q \hookrightarrow A_q$ is injective, since g is monic of positive degree. The algebra has fixed dimension, so arithmetic has constant overhead in prime-field operations. Recursive composition uses this same field and these same constants. □

The lemma is an existence theorem with a uniform algorithm once its finite base data are fixed. It does not give small base cases or usable practical crossover sizes. If effectivity of the base choice is desired, algebraic number coefficients and finite decompositions can be enumerated and their

polynomial identities checked exactly; the existence theorem guarantees termination before the SAT input is processed. Its cost is a fixed constant for a fixed exponent slack. No input-dependent search is hidden inside the asserted exponent.

Proposition 4.3 (Transfer to the contractor ring)**.** *The backend of Lemma 4.1 applies to the matrices over R_p used by the contractor with a polynomial factor in $s+\mathcal{L}$ and the stated input/output storage bound.*

Proof. Use the specified free-module basis of R_p over $\mathbb{Z}/2^p\mathbb{Z}$ and lift each coordinate of each matrix entry to its integer representative in $[0, 2^p)$. For an inner matrix dimension d, an ordinary scalar coordinate product has nonnegative entries bounded by

$$d(2^p - 1)^2.$$

Its bit length is $O(p + \log d)$, polynomial in the root parameter. Compute each such integer matrix product modulo sufficiently many nonexceptional odd primes so that their product exceeds this bound. For every prime, Lemma 4.2 supplies the same finite-base algorithm in A_q; read its scalar output coordinates in $\mathbb{F}_q$. Chinese remaindering then recovers the *integer* coordinate matrix products exactly.

Finally combine the coordinate-pair products with the ring's integer structure constants and reduce modulo 2^p. There are only polynomially many coordinate pairs, and only polynomially many primes of polynomial bit length are needed. They can be processed sequentially, retaining partial CRT outputs of polynomial bit length. Thus both the time overhead and the storage overhead are polynomial.

This argument does not posit a ring homomorphism from R_p to an odd prime field; no such map is used. It first computes ordinary integer products of canonical coordinates and only then restores the ring multiplication. It also never divides by two or by a nonunit of R_p. □

The full continuum of shape bounds in [18] is unnecessary for this theorem. Using a fixed endpoint library is what supplies uniformity for the input-dependent logarithmic dimensions of the actual mode lists.

5 Coupled support certificates

5.1 Complete operation supports and exact selection

A plan partitions the r groups into three ordered nonempty blocks. To build a block, split its ordered list at its midpoint, recursively build both halves, and multiply their boundary tables. Finally contract the three block tables. At an internal merge of disjoint sets A, B with outside set O, the three index classes are AB, AO, BO; the final triangle has the three pair classes between its blocks. There are $r - 3$ internal merges and one final triangle, so every plan has $r - 2$ operations.

If the three class entropy sums are H_1, H_2, H_3, the backend (4.1) is the maximum of nine linear forms on the *unsorted* triple:

$$\mathrm{Perm}(1 + \varepsilon/2, 1 + \varepsilon/2, 0), \qquad \mathrm{Perm}(1, 3/4 + \varepsilon, 1/2). \tag{5.1}$$

This follows from the rearrangement inequality; the first set contains three distinct permutations and the second six. They describe one sorted cost bound, not nine algorithms executed in succession.

For $\varepsilon = 10^{-5}$ a common integer scale is $L_* = 200000$, giving

$$\mathrm{Perm}(200001, 200001, 0), \qquad \mathrm{Perm}(200000, 150002, 100000).$$

Give each unordered group edge its class coefficient and zero if it lies inside one of the parts. For plan j let $\mathcal{V}_j$ be the resulting $9(r-2)$ integer support vectors, and define

$$F_j(g) = L_*^{-1} \max_{v \in \mathcal{V}_j} v \cdot g.$$

Then $2^{F_j(g)}$, up to polynomial factors, pays for every operation of this plan after singleton tables have been formed. The input/output branch of the backend pays for all local boundary arrays, so the same coefficient also bounds local storage.

The algorithm selects the plan of least $F_j(g)$. This comparison can be performed exactly. Put $c_{ab} = |\mathcal{I}_{ab}||\mathcal{I}_{ba}|$ and compare the positive integers

$$P_j = \max_{v \in \mathcal{V}_j} \prod_{a<b} c_{ab}^{v_{ab}}. \tag{5.2}$$

Their bit lengths are polynomial in the root parameter, since the menu, dimension, support scale and exponents are fixed. No floating-point logarithm comparison is necessary. The large fixed scale makes this a complexity specification, not an assertion of practical efficiency.

5.2 An exact hierarchy of quadratic certificates

Theorem 5.1 (Subset-coupled norm certificate)**.** *Let $q \geq 2$ plans have finite nonnegative support families $\mathcal{V}_1, \dots, \mathcal{V}_q$. Give them nonnegative integer weights w_i with $W = \sum_i w_i > 0$. Fix an integer $2 \leq k \leq q$. For each k-element subset $S \subseteq [q]$, let*

$$A_S = \max_{v_i \in \mathcal{V}_i,\ i \in S} \left(\sum_{i \in S} w_i^2 \, \|v_i\|_2^2 + \frac{2(q-1)}{k-1} \sum_{\substack{i<j \\ i,j \in S}} w_i w_j \, v_i \cdot v_j \right). \tag{5.3}$$

Define

$$U_k = \frac{\sum_{|S|=k} A_S}{\binom{q-1}{k-1} W^2}, \qquad \beta_k^2 = \frac{2U_k}{L_*^2 r^2}. \tag{5.4}$$

Selecting the least-cost plan in the menu has local time and local space coefficient at most β_k on the scale $\mathcal{K}(M, u)$.

Proof. For arbitrary choices $v_i \in \mathcal{V}_i$, every diagonal term $w_i^2 \, \|v_i\|_2^2$ occurs in $\binom{q-1}{k-1}$ subsets, and every cross term $w_i w_j v_i \cdot v_j$ in $\binom{q-2}{k-2}$ subsets. Since

$$\frac{q-1}{k-1}\binom{q-2}{k-2} = \binom{q-1}{k-1},$$

we have the exact identity

$$\left\| \sum_i \frac{w_i}{W} v_i \right\|_2^2 = \frac{1}{\binom{q-1}{k-1} W^2} \sum_{|S|=k} \left(\sum_{i \in S} w_i^2 \, \|v_i\|_2^2 + \frac{2(q-1)}{k-1} \sum_{\substack{i<j \\ i,j \in S}} w_i w_j v_i \cdot v_j \right).$$

Replacing each bracket by its independent maximum gives the bound U_k. For the actual entropy vector g, choose a maximizing support v_i for each $F_i(g)$. Then

$$\min_i F_i(g) \leq \sum_i \frac{w_i}{W} F_i(g) = \frac{1}{L_*} \left(\sum_i \frac{w_i}{W} v_i \right) \cdot g \leq (\beta_k + o(1)) \mathcal{K}$$

by (3.4). The menu is finite, so the inherited error is uniform. All local arrays are paid by the input/output support branch, which proves the storage statement as well. □

There is no claim that the separate maximizing tuples for the different subsets are simultaneously realizable. Taking their maxima only relaxes an upper bound. At $k = q$ the certificate retains the complete support tuple; at $k = 2$ it is exactly the incoming pair certificate. The gain from $k = 3$ is proved by the actual integer computations below, not by an assumed strict monotonicity for every arbitrary instance.

5.3 A strict gain with the old backend

Use the incoming nine ten-group plans, weights

$$(137, 124, 135, 120, 108, 119, 106, 87, 64), \qquad W = 1000,$$

and its old integer supports

$$\mathrm{Perm}(1400007, 1400007, 0), \qquad \mathrm{Perm}(1400000, 1175014, 750000), \quad L_* = 1400000.$$

The complete plans are stored as literal data in the archive, with source provenance. There are 72 supports per plan. For $q = 9, k = 3$ the cross coefficient in (5.3) is 8, the normalization is $28W^2$, and the full check visits

$$\binom{9}{3}\, 72^3 = 31,352,832$$

support triples. It gives exactly

$$\beta_{10,\mathrm{old},3}^2 = \frac{13925744235456152241}{42875000000000000000}. \tag{5.5}$$

The incoming pair value was $15952347164548384057/49000000000000000000$. Both use exactly the same operation supports and paid recursion.

Keeping its root taper $\eta_0 = 301/20000$, the new root coefficient is

$$C_{\mathrm{same}} = \kappa \sqrt{\frac{\beta_{10,\mathrm{old},3}^2}{301/20000}} = 29.77681894508374\ldots.$$

The two other recurrence branches remain $29.3352039993294\ldots$ and $29.06574852464125\ldots$, so the root is still the maximum. This proves the same-backend assertion of Theorem 1.1. It isolates a genuine certificate improvement from any improvement due to an imported matrix theorem.

5.4 Updated root certificate and a cyclic thirteen-group menu

With the new supports (5.1), the same weighted triple computation at the root gives

$$\beta_{10}^2 = \frac{16437522499887100933}{56000000000000000000}. \tag{5.6}$$

For thirteen groups, start from the three ordered blocks

$$(0, 1, 2, 4, 7) \mid (3, 6, 8, 12) \mid (5, 9, 10, 11)$$

and add j modulo 13 to every label for $j = 0, \dots, 12$. The resulting 13 plans have equal weights. Their exact pair certificate gives

$$\beta_{13}^2 = \frac{3132736483411}{10985000000000}. \tag{5.7}$$

Each plan is fully charged, including the construction of its blocks. The later menus are finite affine or punctured-affine menus and a final four-group plan, likewise checked with all internal merges included. Their exact data and certificates are included in the standalone verifier.

The seed and schedule were found by exploratory searches. The final proof uses only the displayed rational data, complete finite enumeration, and the analytic bounds above. A search objective or an optimizer's success message is not used as a certificate of correctness or global optimality.

6 A complete paid schedule

6.1 Exact local data

The fixed menus and squared local coefficients for the updated backend are as follows. Pair certificates have equal weights except at the root; the four-group row uses the maximum norm of a single plan's supports.

r	Menu and certificate	β_r^2
10	Nine weighted plans; triples	16437522499887100933/56000000000000000000
13	Thirteen cyclic plans; pairs	3132736483411/10985000000000
9	Four affine-line plans; pairs	31125270001/108000000000
7	Punctured affine plans; pairs	1239384150031/3920000000000
6	Six cyclic plans; pairs	111407262503/337500000000
5	Five cyclic plans; pairs	440004400011/1250000000000
4	$(0,1) \mid (2) \mid (3)$	40000400001/80000000000

The cyclic six-group seed is $(0,1) \mid (2,5) \mid (3,4)$ and the five-group seed is $(0,1) \mid (2,4) \mid (3)$. All shifts modulo the respective group count are used. The nine-point menus are the parallel lines of the affine plane $\mathbb{F}_3^2$; deleting points 7 and 8 gives the seven-group menu with the listed orders retained. Exact literal plans, including the ordered splits inside every block, are part of `data/certificate.json`. The verifier reconstructs their supports; it does not trust stored support vectors or reported decimal optima.

6.2 Tapers and strict packing gates

Let $B = \lceil 6 \log_2 \lambda \rceil$, $u_d = \lfloor \eta_d \lambda \rfloor$, and $a_d = \sum_{j=0}^{d} (u_j + B)$. Use group counts

$$10 \longrightarrow 13 \longrightarrow 9 \longrightarrow 7 \longrightarrow 6 \longrightarrow 5 \longrightarrow 4 \longrightarrow 4 \longrightarrow \cdots$$

and the following rational taper schedule:

$$\begin{aligned} \eta_0 &= \frac{75257}{5000000}, & \eta_1 &= \frac{119723}{10000000}, \quad \eta_2 = \frac{45227}{10000000}, \\ \eta_3 &= \frac{20371}{5000000}, & \eta_4 &= \frac{15323}{5000000}, \quad \eta_5 = \frac{43823}{10000000}, \\ \eta_d &= \frac{641247}{250000000} \left(\frac{63}{100} \right)^{d-6} \quad (d \geq 6). \end{aligned} \tag{6.1}$$

The successive leading cumulative caps, with the last including the entire infinite tail, are

$$(A_0, A_1, A_2, A_3, A_4, A_5, A_\infty) = \frac{(150514, 270237, 315464, 356206, 386852, 430675, 499999)}{10^7}. \quad (6.2)$$

Use relative padding slack λ^{-3}, and stop at $M \le H = \lceil s/\lambda \rceil$ wires.

Integer power comparisons certify the rational upper bounds on $\log_2 r$ in the next table; the last value is exact. The root's dominant exponent is $9/10 + 2A_0 \log_2 10$, whereas later retained-side exponents are $9/10 + A_d \log_2 r_d$, with A_∞ used in the tail.

r_d	Certified bound for $\log_2 r_d$	Gap below one
10	$3321929/10^6$	1589247/2500000000000
13	$3700440/10^6$	104893/250000000000
9	$3169926/10^6$	154021/625000000000
7	$2807355/10^6$	330487/1000000000000
6	$2584963/10^6$	473381/2500000000000
5	$2321929/10^6$	129117/400000000000
4	2	1/5000000

Every gap is strictly positive. The other fresh-side and packet-rounding exponents are bounded by

$$A_\infty \left(1 + 2\frac{3700440}{10^6}\right) < 1.$$

The root's old-side single-Q gate is weaker than its displayed Q^2 gate. Below the root, fresh packet degrees have exponent at most $A_\infty + o(1) < 9/10$. Thus all three conditions in (3.5) hold once s is sufficiently large. The small fixed gaps absorb the additive guards, floors, polylogarithmic factors, and $s/M \le \lambda$. Their smallness is also why these are eventual bounds, not practical finite-size promises.

6.3 Closing the depth and precision induction

To avoid assuming the desired contraction before proving packing, verify the gates provisionally through tail depth $\log_{20/11} \lambda + O(1)$. On this whole range the smallest tail taper has order

$$\lambda^{\delta + o(1)}, \qquad \delta = 1 + \frac{\ln(63/100)}{\ln(20/11)} > 0,$$

because $63/100 > 11/20$. Hence the uniform packing estimates apply through the provisional depth. They give actual tail child ratio $1/2 + o(1) < 11/20$, so an active path cannot exceed that depth. This closes the induction. There are $O(\log \lambda)$ levels and accumulated relative padding error $1 + o(1)$.

At a retained interpolation node, $D \ge 2^{a_{d-1}+B+1}$. The inherited creation-precision bound $O(s\lambda^2 2^{a_{d-1}})$ is at most

$$O(\lambda^4 2^{-B}) = o(1)$$

times MD/u_d, using $s/M \le \lambda$ and $u_d \le \lambda$. Thus the precision and entropy bounds in Section 3 apply uniformly. Moreover $a_d \le A_\infty \lambda + O((\log \lambda)^2)$, so all precisions have polynomial magnitude and polynomial bit length.

The leading wire masses at depths zero through six are

$$(m_0, \dots, m_6) = \left(1, \frac{1}{5}, \frac{2}{65}, \frac{4}{585}, \frac{8}{4095}, \frac{8}{12285}, \frac{16}{61425}\right), \qquad M_d \le (1 + o(1)) m_d s. \quad (6.3)$$

Later masses halve. Original entry routines cost $2^{O(s^{9/10})}$ times a polynomial, new packet routines cost $\exp(s^{A_\infty+o(1)})$ times a polynomial, and direct leaves cost $2^{O(s/\lambda)}$ times their entry overhead. All are lower order on $s/\sqrt{\lambda}$. In the normalized-work recurrence (3.7), multiplying the $\mathrm{poly}(s)$ overheads over $O(\log\lambda)$ levels adds only $O(\lambda\log\lambda)$ to the exponent. Restoring the single factor $P(s+\mathcal{L})$ from Section 3 gives the polynomial description-length factor in the theorem. The imported immutable-entry representation is retained throughout; an entry description is not recursively expanded into an exponentially large expression.

6.4 Exact time and space constants

Define

$$K_d^* = \kappa\frac{m_d}{\sqrt{\eta_d}} \quad (0\le d\le 6), \qquad \gamma = \frac{1/2}{\sqrt{63/100}} = \frac{5}{\sqrt{63}} < 1.$$

At a finite stage $i\le 5$, the aggregation branch of (3.7) has coefficient

$$C_i = \sum_{j<i}\alpha_{r_j}K_j^* + \beta_{r_i}K_i^*.$$

For the tail, $\alpha_4/(1-\gamma) > \beta_4$. Its finite aggregation branches interpolate between the first-tail aggregation and the limiting all-cache route. Consequently the exact global time coefficient is

$$\boxed{C_T = \sum_{j=0}^{5}\alpha_{r_j}K_j^* + \frac{\alpha_4 K_6^*}{1-5/\sqrt{63}}.} \tag{6.4}$$

The finite checker proves that this is larger than every C_i. For orientation, the complete list is

Paid branch	Coefficient
Root aggregation	28.305780030299…
Depth 1 aggregation	28.422653469923…
Depth 2 aggregation	28.155469836586…
Depth 3 aggregation	28.270593027933…
Depth 4 aggregation	28.354261620480…
Depth 5 aggregation	28.380927811887…
Tail, including all recursive cache work	28.433739378341…

Ancestor and descendant tables coexist, so their storage sizes add. There are only $O(\log\lambda)$ active frames, and hence their leading storage coefficient is the maximum of the local coefficients, not their sum. The largest local coefficient is at the root:

$$\boxed{C_S = \beta_{10}K_0^* = \frac{\pi}{\ln 2}\sqrt{\frac{2}{\eta_0}\frac{16437522499887100933}{56000000000000000000}}.} \tag{6.5}$$

The later local coefficients are approximately 6.25666, 1.57435, 0.38609, 0.12996, 0.03741, and 0.02331, and then decay geometrically. Polynomial matrix workspace and CRT overhead do not change these coefficients.

Equations (6.4)–(6.5), with the literal rational data, define the headline values exactly. Rational interval arithmetic proves

$$C_T < 28.434 < 29.81160894538347\ldots, \qquad C_S < 28.306.$$

All finite branches, terminal work, packing and entry costs have now been charged, proving Theorem 1.1. The separate incoming 30.257036…/25.989068… memory point remains useful when its smaller space exponent is preferred.

7 Attempts to improve the logarithmic order

7.1 The effective crossing interface

The Anthropic paper's strongest recurrence is in Appendices A–B, rather than its weaker introductory recurrence. For an alphabet of size 2^μ, it decomposes the crossing tensor

$$\mathsf{C}(x, y, u, v) = [x = v][y = u]$$

as $d\mathsf{C} = \sum_{j \in I} A_j$. Each piece has integer row and column factorizations through c_j^{row} and c_j^{col} bits, with

$$c_j^{\mathrm{row}} + c_j^{\mathrm{col}} \le 4\rho\mu + O(\log \mu). \tag{7.1}$$

There must be polynomially many pieces, polynomial coefficient bit lengths, and a single $2^{O(\mu)}$ evaluation procedure. These effectiveness conditions are essential.

Its parameters satisfy

$$K_{l+1} = g_l^2, \qquad g_l = 2^{3l+x_0} K_l, \qquad e_{l+1} = (\rho + 3\zeta_l)e_l + 3/K_l, \quad \zeta_l = 2^{-l-5}.$$

Therefore $e_l = \Theta((\log K_l)^{-\alpha})$ with $\alpha = \log_2(1/\rho)$. The partition/vertex-operator pieces give $\rho = 1/\sqrt{2}$ and $\alpha = 1/2$ [14].

A successful construction with (7.1) for $\rho < 1/\sqrt{2}$, preserving every stated effectiveness condition, would yield a genuine improvement in the logarithmic denominator power for 3-SAT. The finite probes below do not construct such a family.

7.2 A new aspect-ratio obstruction for the existing two-way bound

An apparently simple way to improve the recursion is to replace its square grid by a rectangle. The following lemma shows precisely why this does not follow from the current total-cost guarantee alone.

Proposition 7.1 (Rectangular-grid minimax)**.** *Let $a, b > 1$, $N = ab$, $0 < \alpha < 1$, $C > 0$, and $0 < \rho < 1$. Suppose the only information about two nonnegative costs is*

$$U_R + U_C \le 2\rho N^2 \mu.$$

Grant the optimistic recursive rates $e(t) = C/(\log_2 t)^\alpha$ and ignore all additional overhead. The worst certified cost of choosing rows or columns is

$$\max_{U_R + U_C \le 2\rho N^2 \mu} \min\{e(a)U_R, e(b)U_C\} = \frac{2\rho C N^2 \mu}{(\log_2 a)^\alpha + (\log_2 b)^\alpha}. \tag{7.2}$$

Among aspect ratios with product N, its minimum is attained by $a = b = \sqrt{N}$. Relative to the target rate $CN^2\mu/(\log_2 N)^\alpha$, the best factor remains $\rho 2^\alpha$.

Proof. For fixed positive multipliers c_R, c_C and budget B, maximizing $\min(c_R U_R, c_C U_C)$ uses all of B and equalizes the two costs. The resulting value is $B/(1/c_R + 1/c_C)$, proving (7.2). Write $x = \log_2 a$, $y = \log_2 b$, so $x + y = \log_2 N$. By concavity,

$$x^\alpha + y^\alpha \le 2\left(\frac{x+y}{2}\right)^\alpha = 2^{1-\alpha}(\log_2 N)^\alpha,$$

with equality at $x = y$. Substitution gives the stated factor. □

Thus the existing two-way guarantee cannot by itself certify progress with $\alpha > \log_2(1/\rho)$; for the current pieces this threshold is $1/2$. This proposition is a minimax statement about an analytic relaxation. It does not assert that every extremal cost pair is realizable by an actual crossing instance, and is not a lower bound for 3-SAT. Extra correlations between row and column costs, or a different crossing identity, could escape its hypotheses.

7.3 Why the straightforward compositions do not close the gap

The newest incoming packets already prove several useful restricted obstructions [10, 12, 13]. We checked them before proposing another version of the same route.

Cauchy modes followed by the clique engine. The explicit labeling layout requires $u \log_2 K = O(\log s)$ for a growing number of groups K. Its information scale is $M/\sqrt{u}$, whereas the current clique coefficient is $\Theta((\log K)^{-1/2})$. Even an ideal, cost-free interface between them would leave the certificate scale

$$\Theta\left(\frac{M}{\sqrt{u \log K}}\right) = \Omega\left(\frac{M}{\sqrt{\log s}}\right).$$

In an actual implementation the local entry routines and precision also have to be paid. This is a limitation of these two certificate scales, not a general runtime lower bound.

Sparse graded spectra in the same energy window. The crossing has row-flattening rank $2^{2\mu}$. If polynomially many pieces factor through polynomially many graded spaces of energies at most $4h$, and $\mu \sim ch^\theta$ while uniformly $\log_2 \dim W_E \le cE^\theta + o(h^\theta)$, rank subadditivity implies

$$2\mu \le O(\log h) + c(4h)^\theta + o(h^\theta), \qquad 2 \le 4^\theta.$$

Hence $\theta \ge 1/2$. Substituting a thinner Hilbert series into the same intermediate-space proof cannot produce the hoped-for improvement. Allowing exponentially many pieces instead makes the existing Cartesian piece enumeration exponential in the original information budget. Structured summation would be a new theorem, not an automatic repair.

Lower initial precision and a conserved degree. The dense reduction has $O(n \log\log n)$ wires and counting precision $n + 1$. Its dominant Hermite-recovery increment is nevertheless $6MD/u$, so lowering the initial precision alone does not change the leading entropy. A representation preserving incidence degree $K = O(n)$ through every recursive interface could instead target $\sqrt{MK/u}$. The prior degree-flow lemma correctly conserves total degree, but does not supply efficient local coefficient queries on the compressed binary indices. Those obligations remain open.

Nonbinary leading digits and separator trees. The incoming odd-prime calculation shows that restricting to one leading digit loses the binary taper cancellation; one fixed nontrivial grouping cannot localize two distinct shifts. Node-dependent groupings are outside that proof but need a new counted layout. The incoming star-weight witness also proves optimality of the flag coefficient among arbitrary separator trees at the used powers of two. Tree imbalance by itself does not improve that polynomial-space certificate.

7.4 Finite rank probes of the actual crossing pieces

The archive implements the Appendix B operator formulas, not merely a dimension surrogate. For $h = 1, \dots, 6$, it computes every piece and checks

$$\sum_{s=-2h}^{2h} A_s = h![x = v][y = u] \pmod{1000003}.$$

For $h = 1, \dots, 8$, the central piece A_0 has the following row and column ranks over that prime field:

h	1	2	3	4	5	6	7	8
$p(h)$	1	2	3	5	7	11	15	22
Row rank	1	4	9	25	49	121	225	484
Column rank	1	4	9	25	49	121	225	484

These integer matrices therefore have full rank over $\mathbb{Q}$ as well. The probes use all $p(h)$ partition labels; the binary theorem selects 2^μ labels, so this statement must not be transferred to every binary submatrix. Also, at these small values the ambient matrix dimension is smaller than the source's intermediate-space upper bound. The experiment does not settle the asymptotic image-rank question. It does reject an unsupported claim of an already visible small-degree rank collapse.

7.5 A further proved local inequality

The supplementary mathematical fragment proves, for sorted $0 \le a \le b \le c$, the symmetric quadratic bound

$$U(a,b,c)^2 \le \frac{55}{96}\left(7(a^2+b^2+c^2) - 4(ab+ac+bc)\right), \tag{7.3}$$

where

$$U(a,b,c) = \begin{cases} b + c, & 2a \le b, \\ c + a + b^2/(4a), & 2a \ge b,\ a > 0. \end{cases}$$

This is the equal-outer rectangular curve followed by blocking, rather than the strongest general three-dimensional bound. Equality in (7.3) occurs at $(1, 1, 20/17)$ and its scaled permutations. An exact affine-plane spectral calculation gives final-triangle envelope coefficient squared $55/192$. The proof and rational checker are included as a separate appendix source.

This last statement concerns three *already formed* blocks. It does not pay for their internal merges or recursive entries, and is not used in C_T or C_S. A fixed-slack algorithmic interpretation would also charge the finite-library approximation slack. The result is a completed local inequality and a possible ingredient for future work, not an additional global SAT improvement.

8 Verification, reproducibility, and limits

8.1 What the finite checks certify

The standalone checker reconstructs every operation support from the literal ordered plans and integer backend coefficients. It recomputes the complete root pair and triple maxima for both backends, all later menu certificates, exact rational packing gaps, cumulative taper sums, child masses, every paid recurrence branch, and the space maximum. The $31,352,832$ root support triples are evaluated for each backend. Pair-dot-product precomputation changes the implementation cost, not which triples are included.

All acceptance comparisons use integers or rational interval arithmetic. Square roots are enclosed using integer arithmetic; logarithm and π intervals have explicit rational remainder bounds. Displayed decimals are produced only after the exact inequalities have passed. The checks run normally and under Python's optimized mode, so removing language-level `assert` statements does not remove a proof gate. Negative controls exercise rejection of corrupted certificate data. The JSON results retain the exact fractions and maxima needed to inspect what was verified.

The separate quadratic checker verifies its polynomial identities and positive Bernstein coefficients, and constructs the affine incidence matrices to verify the exact spectral polynomial and multiplicities. The separate operator probe checks the crossing reconstruction and the finite modular ranks described in Section 7.

8.2 Replaying the archive

From the extracted archive directory, use a Python 3.10-or-newer standard library and run

```
python3 code/run_checks.py
python3 code/build_report.py
```

The first command writes fresh results under `replay_results/`, leaving the delivered result files unchanged. It runs the primary certificate, an independent reconstruction, and the quadratic check. To repeat the separate finite operator experiments as well, NumPy must be available in the selected Python environment; use `python3 code/run_checks.py -full`. The second displayed command invokes `pdflatex` without shell escape and checks the build log for errors, unresolved references, and overfull boxes. It requires a conventional TeX installation with the packages in the preamble. No verification command downloads software, calls GitHub, or mutates a remote repository.

The deliverable includes the complete LaTeX source, compiled PDF, literal certificate data, standalone Python checks, delivered results, a compact source manifest, source-review inventory, and proof-status notes. The source manifest pins the repository and incoming archive identities. It does not duplicate the 34 earlier papers. The ZIP is kept below 4 MB.

8.3 Claims not supplied by these checks

Finite enumeration is not a proof of the inherited analytic contractor, the newer matrix-multiplication theorem, or an end-to-end SAT executable. Those are mathematical dependencies cited in the text. The tests also do not implement the asymptotic matrix algorithms, whose finite base cases may be enormous. This work does not claim a measured SAT speedup, a practical crossover threshold, global optimality of the menus or taper schedule, or a new Lean/Rocq verification of these refinements.

The source review and successful upstream CI evidence make the present status of the Anthropic development much clearer than older reports did. They do not transfer that development's verification status to the new certificate theorem. The small new algebraic statements are reasonable future formalization targets, but no unbuilt Lean file is presented as a checked theorem here.

The remaining research objective is explicit: reduce the general $n \log\log n/\sqrt{\log n}$ exponent order with a complete counted construction. The crossing interface with $\rho < 1/\sqrt{2}$, a recursive compressed degree-preserving representation, or an evaluator that avoids independent mode-index enumeration would be substantive routes. This report provides none of those missing algorithms. Its completed positive result is the strict coefficient improvement with all new finite and recursive costs accounted for.

A A quadratic majorant for rectangular multiplication

The following result concerns the rectangular exponent envelope supplied by $\omega(1,k,1) \le 1+k+1/(4k)$ for $1/2 \le k \le 1$, together with $\omega(1,1/2,1) \le 2$. These shape bounds come from [18]. The finite-algorithm and exact-ring transfer are treated in Section 4; the finite shape library needed here is specified below. The first theorem is an exact algebraic inequality for the displayed function U.

For three nonnegative logarithmic dimensions, sort them as $0 \le a \le b \le c$ and define

$$U(a,b,c) = \begin{cases} b+c, & 2a \le b, \\ c+a+b^2/(4a), & b \le 2a,\ a > 0. \end{cases}$$

For $a = b = 0$ set $U = c$. This is the equal-outer-dimension rectangular curve followed by blocking the largest dimension. It is not asserted to be the strongest general three-dimensional matrix exponent.

Theorem A.1 (Symmetric quadratic majorant)**.** *For all $a,b,c \ge 0$, with U interpreted after sorting,*

$$U(a,b,c)^2 \le \frac{55}{96}\left(7(a^2+b^2+c^2) - 4(ab+ac+bc)\right). \tag{Q}$$

Equality occurs at any positive scalar multiple or permutation of $(1,1,20/17)$. In particular, $55/96$ cannot be reduced while keeping the displayed quadratic form fixed.

Proof. If $b = 0$ the result is immediate. Otherwise divide by b^2 and write $x = a/b$, $y = c/b$. First suppose $1/2 \le x \le 1$, $y \ge 1$. The difference between the right side and $(y + x + 1/(4x))^2$ is $Ay^2 + By + C$, where

$$A = \frac{289}{96}, \qquad B = -\frac{103x^2+55x+12}{24x}, \qquad C = \frac{289x^2-220x+337-6/x^2}{96}.$$

A direct rational calculation gives

$$4AC - B^2 = \frac{(1-x)R(x)}{2304x^2}, \qquad R(x) = -41085x^3 + 67815x^2 - 7590x - 2310.$$

On $[1/2,1]$ the degree-three Bernstein coefficients of R, after $x = (1+t)/2$, are

$$\frac{45705}{8}, \quad 10615, \quad \frac{32065}{2}, \quad 16830.$$

They are strictly positive. Hence $A > 0$ and $4AC - B^2 \ge 0$, which imply $Ay^2 + By + C \ge 0$ for every real y.

For $0 \le x \le 1/2$ the squared left side is $(1+y)^2$. The derivative with respect to x of the difference is

$$\frac{55}{96}(14x - 4 - 4y) < 0 \qquad (y \ge 1).$$

The difference is therefore at least its value at $x = 1/2$, already covered. At $x = 1$, the discriminant vanishes and the quadratic has its zero at $y = 20/17$, proving the equality claim. □

Uniform interpretation with positive slack. Fix $\delta > 0$. Choose a finite rational mesh in $[1/2, 1]$, containing both endpoints, and a positive slack for each of the finitely many algorithms from [18]. For $1/2 \le a/b \le 1$, round a/b up to the next mesh point k, pad the inner dimension, and block the largest dimension. The exponent is at most

$$c + b\left(k + \frac{1}{4k} + \eta\right).$$

The derivative of $k + 1/(4k)$ lies in $[0, 3/4]$ on this interval. Thus a sufficiently fine fixed mesh and sufficiently small fixed $\eta > 0$ bound the padding and algorithmic loss by $\delta(a+b+c)$. For $a/b < 1/2$, use the half-width endpoint. Rounding finite-base recursion counts contributes only fixed dimension factors. The mesh can be selected using exact integer-power comparisons of the input dimensions, so no real-number oracle is required. Proposition 4.3 applies to this finite collection after taking one common number field and the union of its finite exceptional-prime sets.

Consequently every fixed positive enlargement of a coefficient derived from U is realized by a fixed finite library. The exact envelope and its limiting coefficient do not assert attainment of an exponent infimum with zero slack. The main time and space coefficients use the separate fixed-slack endpoint library of Lemma 4.1.

B A final-triangle certificate from the affine plane of order three

Identify the nine group labels with $\mathbb{F}_3^2$. Its four directions partition the points into three parallel lines. For direction j, let H_j be the three-by-36 zero-one matrix whose rows indicate the three unordered pairs of line classes. For a nonnegative unordered-edge entropy vector $g \in \mathbb{R}^{36}_{\ge 0}$, its three aggregate dimensions are $H_j g$.

Theorem B.1 (Final triangle only)**.** *There is a direction j with*

$$U(H_j g) \le \sqrt{\frac{1485}{128}}\, \|g\|_2. \tag{A}$$

In the inherited nine-group mode-entropy normalization, this gives the local final-triangle coefficient

$$\beta^2_{9,\mathrm{final}} = \frac{55}{192}, \qquad \beta_{9,\mathrm{final}} = 0.5352180241\ldots.$$

This statement does not pay for producing the three line-block tensors.

Proof. Stack the four H_j into a twelve-by-36 matrix H. Each row has nine ones. Within a direction the rows have disjoint supports; between two directions any two rows have inner product two. The latter count is obtained from the four intersections of two selected lines in each direction: exactly the two opposite corner pairs belong to both supports. Thus HH^T has diagonal blocks $9I_3$ and off-diagonal blocks $2J_3$. Let K be block diagonal with four copies of $9I_3 - 2J_3$. By (Q),

$$\frac{1}{4}\sum_{j=1}^{4} U(H_j g)^2 \le g^T \left(\frac{55}{384} H^T K H\right) g.$$

The nonzero eigenvalues of $H^T K H$ are 81 with multiplicity nine and 9 with multiplicity three. To see this directly, decompose the 12-dimensional row space into the eight-dimensional subspace having coordinate sum zero in each block, the three-dimensional subspace of blockwise constants whose total is zero, and the one-dimensional constant subspace. On these, HH^T acts by $9, 3, 27$,

respectively, while K acts by $9, 3, 3$. Their product therefore acts by $81, 9, 81$. The largest eigenvalue of the averaged quadratic form is $55 \cdot 81/384 = 1485/128$. The minimum of four nonnegative numbers is at most their root-mean-square, proving (A).

If the directed mode entropy vector has Euclidean norm at most $(1 + o(1))\mathcal{K}/9$, then joining the two orientations gives $\|g\|_2 \le (\sqrt{2}/9 + o(1))\mathcal{K}$. Substitution into (A) yields $\beta^2 = 2(1485/128)/81 = 55/192$. □

Scope of the result. The proof covers the final aggregation of three precomputed blocks. To interpret it algorithmically with a fixed finite library, replace $\beta_{9,\mathrm{final}}$ by any strictly larger coefficient, as explained above. It gives no bound for producing those block tensors, for a complete contraction plan, or for the full 3-SAT recurrence. The internal line-block merges and recursive cache construction must still be charged. No verified combination that improves the main fully paid coefficient is established here, and this appendix's coefficient is not used in that calculation. A common quadratic certificate for both internal and final operations, or an adaptive combination of linear and quadratic certificates, remains a possible continuation of this local result.

References

[1] OpenAI, *The Exponential Time Hypothesis is False*, revised 25 October 2026. Quantitative reduction and contraction appendix. Upstream revision `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`. https://github.com/openai/math/tree/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026. The ProveIt vendored LaTeX was inspected byte for byte in the frozen snapshot.

[2] OpenAI, formalization scope for result 378, https://github.com/openai/math/blob/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/lean/docs/378.md.

[3] Vladimir Reshetnikov, ProveIt, *Sub-exponential 3-SAT* project index, revision `8e76d515063c4b9fbd8743a846b457e3a01bc039`. https://github.com/VladimirReshetnikov/ProveIt/blob/8e76d515063c4b9fbd8743a846b457e3a01bc039/Computability/SubexponentialSAT/README.md.

[4] ProveIt, incoming archive commit `26bd83d4b37d3c5a3ccb45bab2e6385fae3977f1`, https://github.com/VladimirReshetnikov/ProveIt/commit/26bd83d4b37d3c5a3ccb45bab2e6385fae3977f1. The five archives are identified by Git blob and SHA-256 in the accompanying source manifest.

[5] *3-SAT precision refinement*, 26 October 2026. ProveIt, `Research/3sat_precision_refinement/report.tex` under `Computability/SubexponentialSAT`, at the frozen revision. Exact Hermite losses and optimal-taper analysis.

[6] *Asymmetric packet balancing*, 26 October 2026. ProveIt, `Research/eth_asymmetric_packet_balancing/report.tex` at the frozen revision. Near-unit padding and asymmetric degree tracking.

[7] *Joint Cauchy modes*, 26 October 2026. ProveIt, `Research/eth_joint_cauchy/report.tex`, together with `Research/sat_adaptive_bounds/report/interface.tex`, at the frozen revision. Full-orientation mode weights and entropy interface.

[8] *Adaptive bounds for exact SAT contraction*, 26 October 2026. ProveIt, `Research/sat_adaptive_bounds/` at the frozen revision. The generic polynomial-space coefficient and the shared weighted entropy and matrix-ring interfaces.

[9] *Adaptive contraction-tree certificates*, 26 October 2026. ProveIt, `Research/sat_adaptive_tree_certificate/` at the frozen revision. Complete operation supports and cached recurrence accounting.

[10] *Pairwise certificates for exact SAT contraction*, `sat_pairwise_certificates_2026-10-26.zip`, from [4]. Sections `interface`, `portfolio`, `schedule`, `barriers`, and `research`. The immediate comparison baseline and the suggested higher-order certificate route.

[11] *Affine-plane contraction plans for exact SAT*, `sat_affine_contraction_2026-10-26.zip`, from [4]. Joint time–space selector and the smaller-memory comparison point.

[12] *Affine-plane refinement*, `sat_affine_plane_2026-10-26.zip`, from [4]. Sections `affine` and `obstructions`.

[13] *Affine Cauchy refinement*, `sat_affine_cauchy_2026-10-26.zip`, from [4]. The nested-clique and sparse-spectrum research analysis.

[14] Claude (Anthropic), *A $2^{o(n)}$ time algorithm for 3-SAT*, `eth/paper.pdf` at commit `1a6eee293b9e00cb432eef6018a21f523623a9af`. https://github.com/anthropics/formal-math/blob/1a6eee293b9e00cb432eef6018a21f523623a9af/eth/paper.pdf. Appendices A and B supply the effective crossing interface and operators.

[15] Anthropic, public formal-math development, `eth/MainResults.lean` and `eth/Solution.lean`, same revision. https://github.com/anthropics/formal-math/tree/1a6eee293b9e00cb432eef6018a21f523623a9af/eth.

[16] Anthropic, GitHub Actions `verify (eth)` job, exact-commit verification log, retrieved 26 October 2026 (Pacific time). https://github.com/anthropics/formal-math/actions/runs/40440389861/job/121979112896.

[17] OpenAI, *Matrix multiplication with exponent at most* 9/4, formalization scope for result 107, https://github.com/openai/math/blob/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/lean/docs/107.md. The accompanying comparator statements distinguish finite arithmetic program exponents from bit complexity.

[18] Przemysław Uznański, *Rectangular matrix multiplication from polynomial products*, arXiv:2610.10494v2, 11 October 2026. https://arxiv.org/html/2610.10494v2. Theorem 1.1, Corollary 1.2, characteristic-zero convention, and the finite-to-asymptotic algorithm discussion.